

Classification Algorithms

Dataset: All Electronics Customer Database

Examples

1. Decision Tree

a) ID3

b) CART

c) C4.5

2. Naive Bayesian

3. K-Nearest Neighbour.

Classification Algorithms

Def: A classification Algorithm is a procedure for selecting a hypothesis from a set of alternatives that best fits a set of observations.

* It is a supervised learning i.e. it has a specific class/target variable.

Algorithms

1. Decision Tree $\begin{cases} ID_3 \\ CART \\ C4.5 \end{cases}$
2. Naive Bayesian — based on probability (measures)
3. K-Nearest Neighbour $\begin{cases} \text{categorical} \rightarrow \text{simple similarity Matching} \\ \text{Numerical} \rightarrow \text{Euclidean distance} \end{cases}$

Decision tree Algorithm	Attribute Measure	Formulas
ID ₃ (non-binary tree)	InfoGain	$1. \text{Info}(D) = - \sum_{i=1}^m P_i \log_2 P_i$ $2. \text{Info}_A(D) = \sum_{j=1}^v \frac{ D_j }{ D } \text{Info}(D_j)$ $3. \text{Inf Gain}_A(D) = \text{Info}(D) - \text{Info}_A(D)$
CART (binary tree)	Gini Index	$1. \text{Gini}(D) = 1 - \sum_{i=1}^m P_i^2$ $2. \text{Gini}_A(D) = \frac{ D_1 }{ D } \text{Gini}(D_1) + \frac{ D_2 }{ D } \text{Gini}(D_2)$ $3. \text{Gini}(A) = \text{Gini}(D) - \text{Gini}_A(D)$
C4.5 Extension of ID ₃ (Non-binary tree)	Gain Ratio	$1. \text{Gain}(D) = \text{Info}(D) - \text{Info}_A(D)$ $2. \text{SplitInfo}_A(D) = - \sum_{j=1}^v \frac{ D_j }{ D } \log_2 \frac{ D_j }{ D }$ $3. \text{GainRatio}_A(D) = \frac{\text{Gain}(D)}{\text{SplitInfo}_A(D)}$

Dataset: AllElectronics Customer Database

RID	age	Income	Student	Credit-Rating	class: buys-computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-aged	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-aged	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-aged	Medium	No	Excellent	Yes
13	Middle-aged	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

ID3 - Iterative Dichotomiser 3

* It is measured by the Concept of Information gain

Formulas:

1. The expected information needed to classify a tuple in D is given by

$$\text{Info}(D) = - \sum_{i=1}^m P_i \log_2(P_i)$$

Where,

P_i = nonzero probability

m = no. of classes.

$\text{Info}(D)$ is also known as Entropy of D .

2. Now, to partition the tuples in D on some attribute A

$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

Where,

v = no. of categories in an attribute

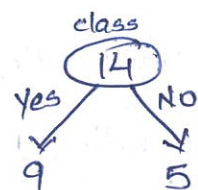
3. Information gain is defined as the difference between the Original information requirement (i.e. based on classes) and new requirement (i.e., obtained after partitioning on A).

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

Problem: Consider AllElectronics Customer Database

step1: Compute Original Info. requirement for class i.e. Yes, No

$$\begin{aligned} \text{Info}(D) &= - \sum_{i=1}^m P_i \log_2(P_i) \\ &= - \frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ &= \boxed{0.940} \end{aligned}$$



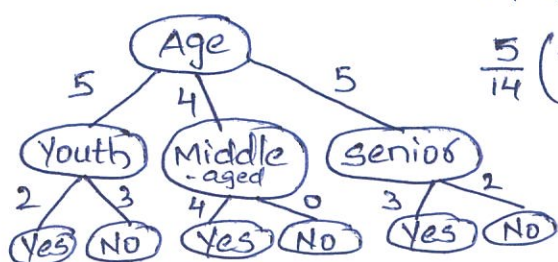
(4)

Step 2:

Compute the expected information requirement for each attribute i.e. for age, Income, student, credit-rating.

$$\text{Info}_A(D) = \sum_{j=1}^n \frac{|D_j|}{|D|} \text{Info}(D_j)$$

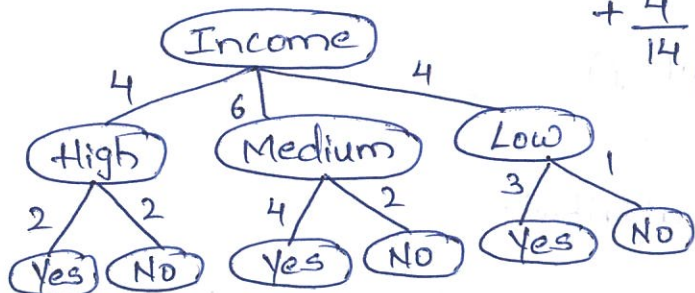
$$(i) \text{Info}_{\text{Age}}(D) = \frac{5}{14} \left(-\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right) + \frac{4}{14} \left(-\frac{4}{4} \log_2 \left(\frac{4}{4} \right) - 0 \right) + \frac{5}{14} \left(-\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$



$$= 0.346 + 0 + 0.346$$

$$= \boxed{0.692}$$

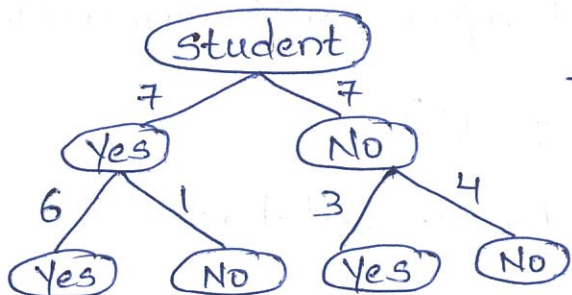
$$(ii) \text{Info}_{\text{Income}}(D) = \frac{4}{14} \left[-\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right] + \frac{6}{14} \left[-\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right] + \frac{4}{14} \left[-\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right]$$



$$= 0.285 + 0.393 + 0.231$$

$$= \boxed{0.909}$$

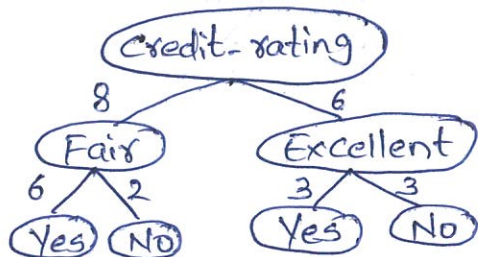
$$(iii) \text{Info}_{\text{Student}}(D) = \frac{7}{14} \left[-\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right] + \frac{7}{14} \left[-\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right]$$



$$= 0.295 + 0.492$$

$$= \boxed{0.787}$$

$$(iv) \text{Info}_{\text{Credit-rating}}(D) = \frac{8}{14} \left[-\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) \right] + \frac{6}{14} \left[-\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right]$$



$$= 0.463 + 0.428$$

$$= \boxed{0.891}$$

Step 3: Now, Compute Gain for each attribute.

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

$$(i) \text{Gain}(\text{age}) = 0.940 - 0.692 = \boxed{0.248} \text{ Maximum}$$

$$(ii) \text{Gain}(\text{Income}) = 0.940 - 0.909 = 0.031$$

$$(iii) \text{Gain}(\text{student}) = 0.940 - 0.787 = 0.153$$

$$(iv) \text{Gain}(\text{Credit-Rating}) = 0.940 - 0.891 = 0.049$$

* Now, pick the attribute that has the highest information gain.

* Here "age" is having the highest gain so consider age as a root node (i.e. splitting attribute)

* The tuples of age should be partitioned.

Youth				Senior			
Income	student	C-R	buys-Computers	Income	student	C-R	buys-Computers
High	No	Fair	No	Medium	No	Fair	Yes
High	No	Excellent	No	low	Yes	Fair	Yes
Medium	No	Fair	No	low	Yes	Excellent	No
Low	Yes	Fair	Yes	Medium	Yes	Fair	Yes
Medium	Yes	Excellent	Yes	Medium	No	Excellent	No

Middle-aged			
Income	student	Credit-Rating	buys-Computers
High	No	Fair	Yes
low	Yes	Excellent	Yes
Medium	No	Excellent	Yes
High	Yes	Fair	Yes

* For age = middle-aged all belong to the same class i.e. Yes.
Hence, a leaf node should be created at the end of this branch and labeled as "Yes"

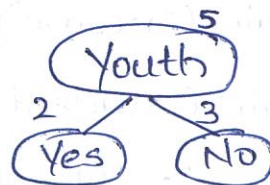
* But for the remaining branches i.e. age = Youth, Senior the above 3 steps must be computed until leaf nodes are created.

Consider a branch = Youth (D)

Step 1:

$$\text{Info}(D) = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right)$$

$$= \boxed{0.970}$$

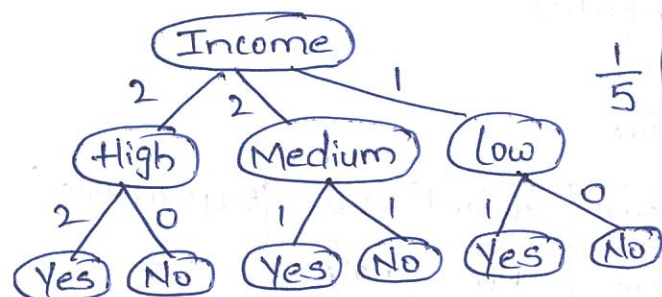


Step 2:

$$(i) \text{Info}_{\text{income}}(D) = \frac{2}{5} \left[-\frac{2}{2} \log_2\left(\frac{2}{2}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right] + \frac{1}{5} \left[-\frac{1}{1} \log_2\left(\frac{1}{1}\right) \right]$$

$$= 0 + 0.4 + 0$$

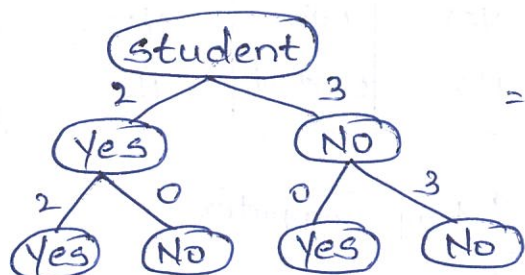
$$= 0.4$$



$$(ii) \text{Info}_{\text{student}}(D) = \frac{2}{5} \left[-\frac{2}{2} \log_2\left(\frac{2}{2}\right) \right] + \frac{3}{5} \left[-\frac{3}{3} \log_2\left(\frac{3}{3}\right) + 0 \right]$$

$$= 0 + 0$$

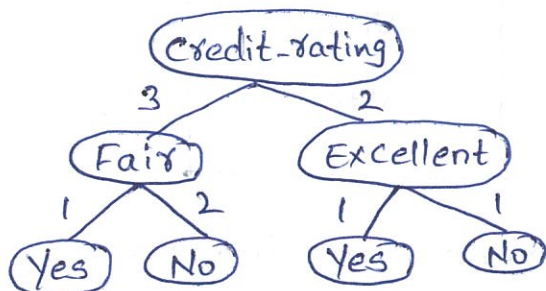
$$= 0$$



$$(iii) \text{Info}_{\text{credit-rating}}(D) = \frac{3}{5} \left[-\frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$

$$= 0.550 + 0.4$$

$$= 0.95$$



Step: 3

$$i) \text{Gain}(\text{Income}) = 0.970 - 0.4 = 0.57$$

$$ii) \text{Gain}(\text{Student}) = 0.970 - 0 = \boxed{0.97} \text{ Maximum}$$

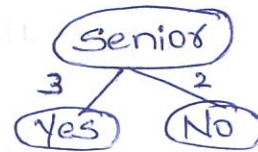
$$iii) \text{Gain}(\text{Credit-Rating}) = 0.970 - 0.95 = 0.02$$

Now, Compute $\text{age} = \text{senior}(D)$

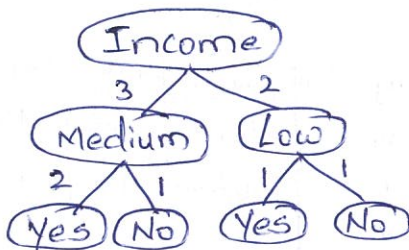
Step 1:

$$\text{Info}(D) = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$

$$= \boxed{0.970}$$

Step 2:

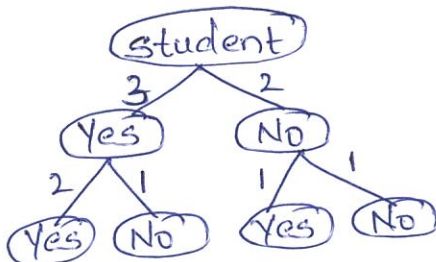
$$i) \text{Info}_{\text{income}}(D) = \frac{3}{5} \left[-\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$



$$= 0.550 + 0.4$$

$$= \boxed{0.95}$$

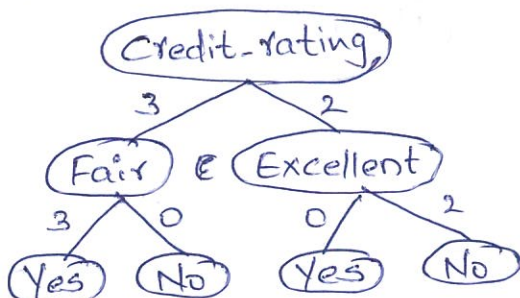
$$ii) \text{Info}_{\text{student}}(D) = \frac{3}{5} \left[-\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$



$$= 0.550 + 0.4$$

$$= \boxed{0.95}$$

$$iii) \text{Info}_{\text{credit-rating}}(D) = \frac{3}{5} \left[-\frac{3}{3} \log_2\left(\frac{3}{3}\right) - 0 \right] + \frac{2}{5} \left[0 - \frac{2}{2} \log_2\left(\frac{2}{2}\right) \right]$$

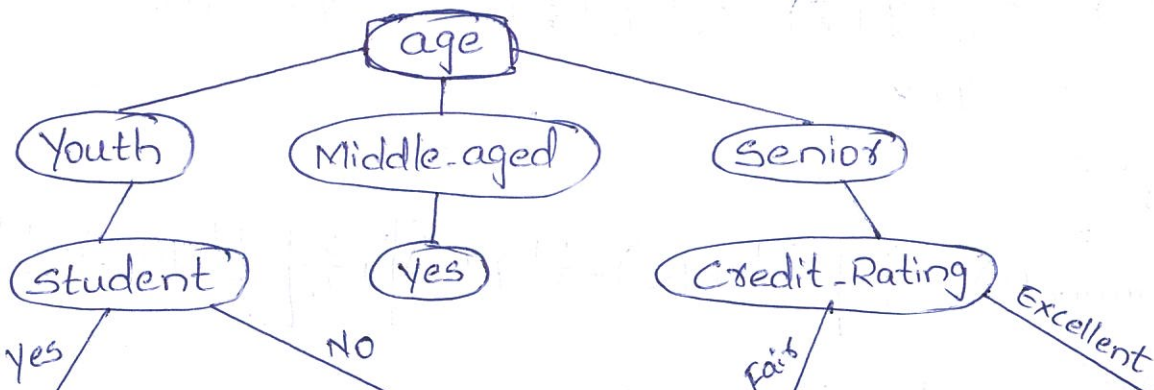


$$= \boxed{0}$$

Step: 3

- (i) $\text{Gain}(\text{Income}) = 0.970 - 0.95 = 0.02$
 (ii) $\text{Gain}(\text{Student}) = 0.970 - 0.95 = 0.02$
 (iii) $\text{Gain}(\text{Credit-Rating}) = 0.970 - 0 = \boxed{0.970} \text{ Max.}$

* For, age = Youth, the highest gain attribute is student
 For, age = Senior, the highest gain attribute is credit-rating.



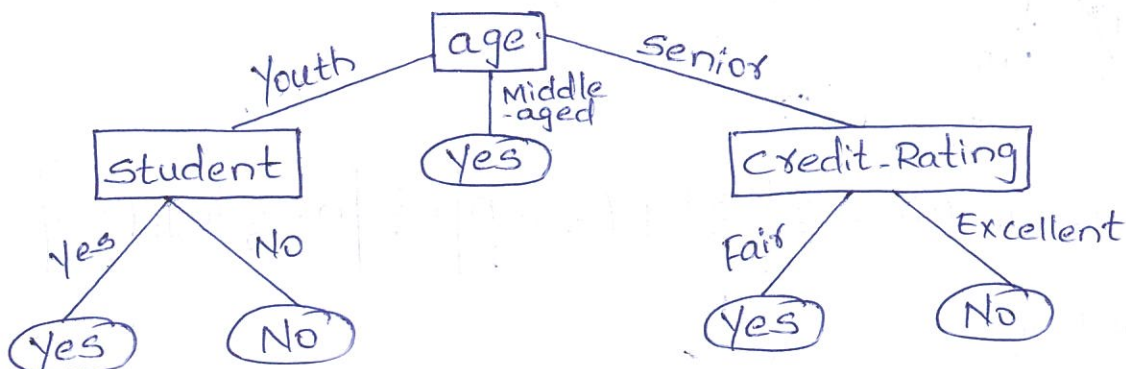
Income	C-R	buys.comp
Low	Fair	Yes
Medium	Excellent	Yes

Income	C-R	buys-comp
High	Fair	No
High	Excellent	No
High	Fair	No

Income	student	buys-comp
Medium	No	Yes
Low	Yes	Yes
Medium	Yes	Yes

Income	student	buys-co
Low	Yes	No
Medium	No	No

- * The branches of student and credit-rating are all belong to same class. Hence leaf nodes should be created.
 * The final decision tree is as follows.



CART - Classification and Regression Trees

- * It is the generation of binary trees
- * The attribute measure used is Gini Index

Formulas

1. To measure the impurity of Dataset (D), data partition or set of training tuples as

$$\text{Gini}(D) = 1 - \sum_{i=1}^m p_i^2$$

2. When considering a binary split, we compute sum of each partition as follows

$$\text{Gini}_A(D) = \frac{|D_1|}{|D|} \text{Gini}(D_1) + \frac{|D_2|}{|D|} \text{Gini}(D_2)$$

Note: $2^n - 2$ possible ways to form two partitions of the data.

Where, n = no. of Categories in an attribute

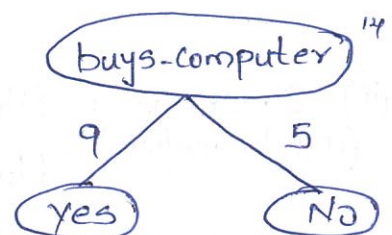
3. The reduction in impurity that would be incurred by a binary split is

$$\text{Gini}(A) = \text{Gini}(D) - \text{Gini}_A(D)$$

Problem: Consider Allelectronics Customer Database.

step 1: Compute the impurity of D.

$$\begin{aligned} \text{Gini}(D) &= 1 - \sum_{i=1}^m p_i^2 \\ &= 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 \\ &= 1 - [0.4133 + 0.1275] \\ &= \boxed{0.459} \end{aligned}$$



step 2: Find the splitting criterion for each attribute i.e. age, Income, student, Credit-rating.

- (i) Age = Youth, middle-aged, senior $(2^n - 2) = 2^3 - 2 = 6$
 $\{Youth\}, \{Middle-aged\}, \{senior\}, \{Youth, middle-aged\}, \{Youth, senior\}, \{middle-aged, senior\}$

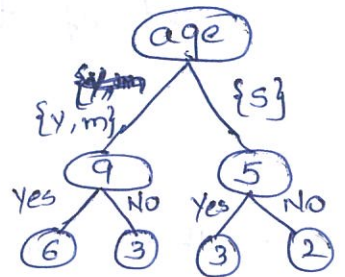
* Gini_{age} $\in \{Youth, Middle-aged\} (D)$

$$= \frac{9}{14} \left[1 - \left(\left(\frac{6}{9} \right)^2 + \left(\frac{3}{9} \right)^2 \right) \right] + \frac{5}{14} \left[1 - \left(\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right) \right]$$

$$= \frac{9}{14} (0.444) + \frac{5}{14} (0.48)$$

$$= 0.285 + 0.171$$

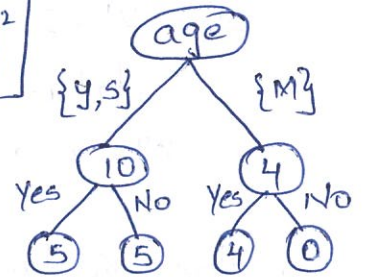
$$= \boxed{0.456}$$



* Gini_{age} $\in \{y, s\} (D)$ $= \frac{10}{14} \left[1 - \left(\left(\frac{5}{10} \right)^2 + \left(\frac{5}{10} \right)^2 \right) \right] + \frac{4}{14} \left[1 - \left(\frac{4}{4} \right)^2 \right]$

$$= \frac{10}{14} (0.5) + \frac{4}{14} (0)$$

$$= \boxed{0.357} \text{ Min.}$$

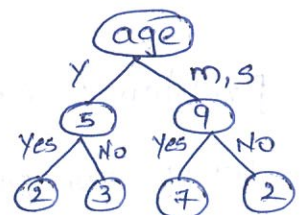


* Gini_{age} $\in \{m, s\} (D)$ $= \frac{9}{14} \left[1 - \left(\left(\frac{7}{9} \right)^2 + \left(\frac{2}{9} \right)^2 \right) \right] + \frac{5}{14} \left[1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) \right]$

$$= \frac{9}{14} (0.346) + \frac{5}{14} (0.48)$$

$$= 0.222 + 0.171$$

$$= \boxed{0.393}$$

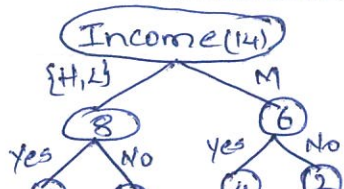


- (ii) Income = {High}, {low}, {medium}, {High, low}, {High, medium}, {High, low, medium}, {low, medium} $(2^n - 2) = 2^3 - 2 = 6$

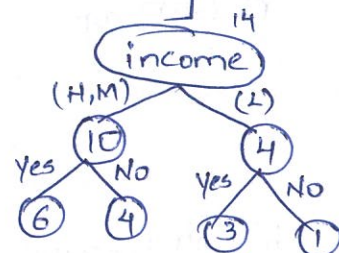
* Gini_{income} $\in \{H, L\} (D)$ $= \frac{8}{14} \left[1 - \left(\left(\frac{5}{8} \right)^2 + \left(\frac{3}{8} \right)^2 \right) \right] + \frac{6}{14} \left[1 - \left(\left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right) \right]$

$$= \frac{8}{14} (0.469) + \frac{6}{14} (0.444)$$

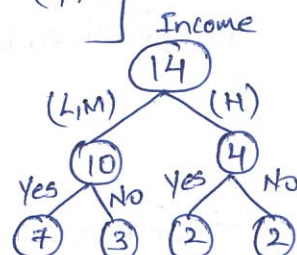
$$= 0.268 + 0.190 = \boxed{0.458}$$



$$\begin{aligned}
 * \text{Gini}_{\text{income}} \in \{H, M\} (D) &= \frac{10}{14} \left[1 - \left(\left(\frac{6}{10} \right)^2 - \left(\frac{4}{10} \right)^2 \right) \right] + \frac{4}{14} \left[1 - \left(\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right) \right] \\
 &= \frac{10}{14} (0.48) + \frac{4}{14} (0.375) \\
 &= 0.342 + 0.107 \\
 &= \boxed{0.449}
 \end{aligned}$$

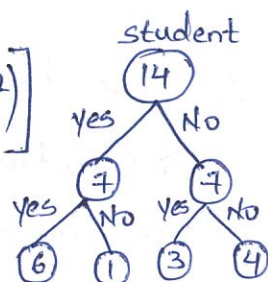


$$\begin{aligned}
 * \text{Gini}_{\text{income}} \in \{L, M\} (D) &= \frac{10}{14} \left[1 - \left(\left(\frac{7}{10} \right)^2 + \left(\frac{3}{10} \right)^2 \right) \right] + \frac{4}{14} \left[1 - \left(\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right) \right] \\
 &= \frac{10}{14} (0.42) + \frac{4}{14} (0.5) \\
 &= 0.3 + 0.142 \\
 &= \boxed{0.442} \text{ Min.}
 \end{aligned}$$



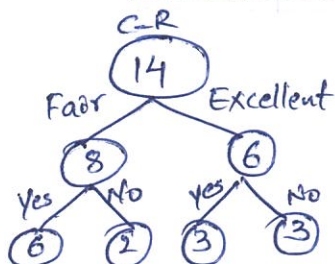
iii) Student = Yes, No ; $2^n - 2 \Rightarrow 2^2 - 2 = 2$

$$\begin{aligned}
 * \text{Gini}_{\text{Student}} \in \{\text{Yes}\} (D) &= \frac{7}{14} \left[1 - \left(\left(\frac{6}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right) \right] + \frac{7}{14} \left[1 - \left(\left(\frac{3}{7} \right)^2 + \left(\frac{4}{7} \right)^2 \right) \right] \\
 &= \frac{7}{14} (0.245) + \frac{7}{14} (0.49) \\
 &= 0.122 + 0.245 \\
 &= \boxed{0.367}
 \end{aligned}$$



iv) Credit-rating = Fair, Excellent = $2^n - 2 = 2^2 - 2 = 2$
 $\Rightarrow \{\text{Fair}\}, \{\text{Excellent}\}$

$$\begin{aligned}
 * \text{Gini}_{\text{Credit-rating}} \in \{\text{Fair}\} (D) &= \frac{8}{14} \left[1 - \left(\left(\frac{6}{8} \right)^2 + \left(\frac{2}{8} \right)^2 \right) \right] + \frac{6}{14} \left[1 - \left(\left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right) \right] \\
 &= \frac{8}{14} (0.375) + \frac{6}{14} (0.5) \\
 &= 0.214 + 0.214 \\
 &= \boxed{0.428}
 \end{aligned}$$



Now, Consider minimum Gini value of splitting subset of each attribute.

- i) $\text{age} \in \{\text{Youth}, \text{senior}\} = 0.357$
- ii) $\text{Income} \in \{\text{Low}, \text{Medium}\} = 0.442$
- iii) $\text{Student} \in \{\text{Yes}\} = 0.367$
- iv) $\text{Credit-rating} \in \{\text{Fair}\} = 0.428$

Step 3: Compute Gini Index

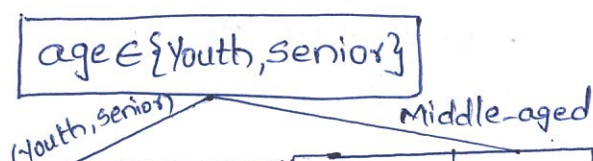
$$\begin{aligned} \text{i) } Gini_{age} &= Gini(D) - Gini_{age}(\{Y, S\}) \\ &= 0.459 - 0.357 \\ &= \boxed{0.102} \text{ Maximum} \end{aligned}$$

$$\begin{aligned} \text{ii) } Gini_{Income} &= Gini(D) - Gini_{Income}(\{L, M\}) \\ &= 0.459 - 0.442 = \boxed{0.017} \end{aligned}$$

$$\begin{aligned} \text{iii) } Gini_{student} &= Gini(D) - Gini_{student}(Yes) \\ &= 0.459 - 0.367 \\ &= \boxed{0.092} \end{aligned}$$

$$\begin{aligned} \text{iv) } Gini_{C-R} &= Gini(D) - Gini_{C-R}(Fair) \\ &= 0.459 - 0.428 \\ &= \boxed{0.031} \end{aligned}$$

∴ The binary split, $age = \{Youth, Senior\}$ results in maximum reduction in impurity of tuples in D. so Consider "age" as a root node.



age	Income	student	C-R	buys-Comp	age	Income	student	C-R	buys-Comp
Youth	High	No	Fair	No	Middle-aged	High	No	Fair	Yes
Youth	High	No	Excellent	No	Middle-aged	Low	Yes	Excellent	Yes
Senior	Medium	No	Fair	Yes	Middle-aged	Medium	No	Excellent	Yes
Senior	Low	Yes	Fair	Yes	Middle-aged	High	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No					
Youth	Medium	No	Fair	No					
Youth	Low	Yes	Fair	Yes					
Senior	Medium	Yes	Fair	Yes					
Youth	Medium	Yes	Excellent	Yes					
Senior	Medium	No	Excellent	No					

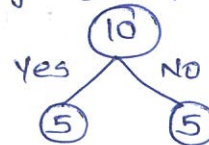
* For $age \in \text{Middle-aged}$ all belong to the same class "yes" hence leaf node should be created.

Consider a branch, $age \in \{\text{Youth}, \text{Senior}\}$ and follow the same Procedure until leaf nodes are created.

Step 1:

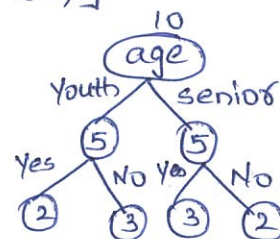
$$\begin{aligned} \text{Gini}(D) &= \left[1 - \left(\left(\frac{5}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right) \right] \\ &= (1 - 0.255) \\ &= \boxed{0.745} \end{aligned}$$

$age \in \{\text{Youth}, \text{Senior}\}$

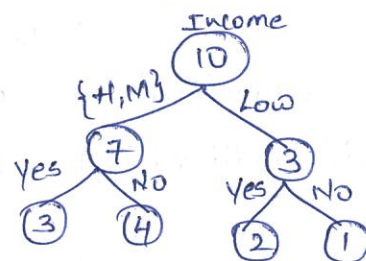


Step 2:

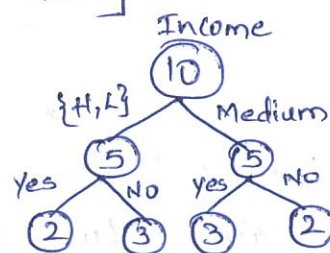
$$\begin{aligned} \text{(i) } \text{Gini}_{age \in \text{Youth}} &= \frac{5}{10} \left[1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) \right] + \frac{5}{10} \left[1 - \left(\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right) \right] \\ &= \frac{5}{10} (0.48) + \frac{5}{10} (0.48) \\ &= 0.24 + 0.24 \\ &= \boxed{0.48} \end{aligned}$$



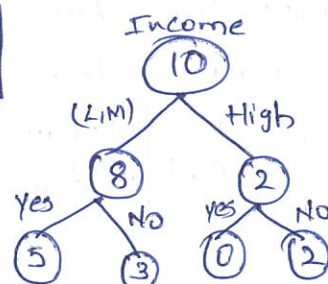
$$\begin{aligned} \text{(ii) } * \text{Gini}_{income \in \{H, M\}} &= \frac{7}{10} \left[1 - \left(\left(\frac{3}{7} \right)^2 + \left(\frac{4}{7} \right)^2 \right) \right] + \frac{3}{10} \left[1 - \left(\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right) \right] \\ &= \frac{7}{10} (0.49) + \frac{3}{10} (0.444) \\ &= 0.343 + 0.133 \\ &= \boxed{0.476} \end{aligned}$$



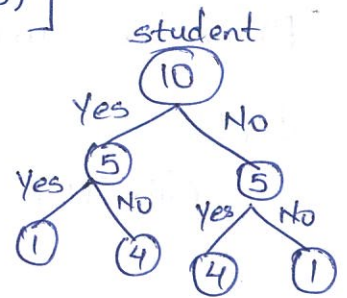
$$\begin{aligned} * \text{Gini}_{income \in \{H, L\}} &= \frac{5}{10} \left[1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) \right] + \frac{5}{10} \left[1 - \left(\left(\frac{3}{5} \right)^2 + \left(\frac{2}{5} \right)^2 \right) \right] \\ &= \frac{5}{10} (0.48) + \frac{5}{10} (0.48) \\ &= 0.24 + 0.24 \\ &= \boxed{0.48} \end{aligned}$$



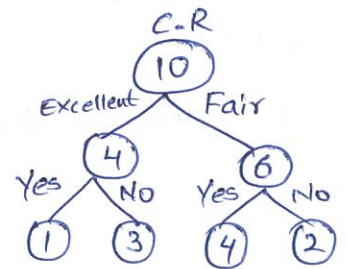
$$\begin{aligned} * \text{Gini}_{income \in \{L, M\}} &= \frac{8}{10} \left[1 - \left(\left(\frac{5}{8} \right)^2 + \left(\frac{3}{8} \right)^2 \right) \right] + \frac{2}{10} \left[1 - \left(\left(\frac{2}{2} \right)^2 \right) \right] \\ &= \frac{8}{10} [1 - 0.531] \\ &= \frac{8}{10} (0.469) \\ &= \boxed{0.375} \text{ Min} \end{aligned}$$



$$\begin{aligned}
 \text{iii) } \underline{Gini_{\text{student}} \in \text{No}} &= \frac{5}{10} \left[1 - \left(\frac{1}{5} \right)^2 + \left(\frac{4}{5} \right)^2 \right] + \frac{5}{10} \left[1 - \left(\frac{4}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] \\
 &= \frac{5}{10} (0.32) + \frac{5}{10} (0.32) \\
 &= 0.16 + 0.16 \\
 &= \boxed{0.32}
 \end{aligned}$$



$$\begin{aligned}
 \text{iv) } \underline{Gini_{C-R} \in \text{Excellent}} &= \frac{4}{10} \left[1 - \left(\frac{1}{4} \right)^2 + \left(\frac{3}{4} \right)^2 \right] + \frac{6}{10} \left[1 - \left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right] \\
 &= \frac{4}{10} (0.375) + \frac{6}{10} (0.444) \\
 &= 0.15 + 0.26 \\
 &= \boxed{0.41}
 \end{aligned}$$



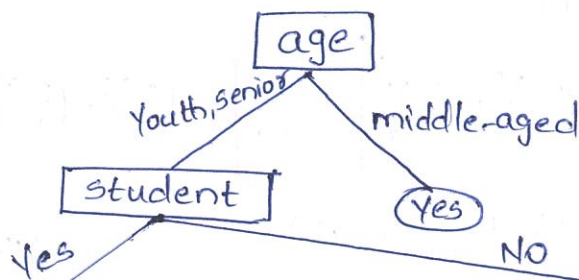
Step 3:

$$Gini(\text{age}) = 0.745 - 0.48 = \underline{0.265}$$

$$\begin{aligned}
 Gini(\text{Income}) &= Gini(0) - Gini_{\text{income}} \{L, M\} \\
 &= 0.745 - 0.375 = \underline{0.37}
 \end{aligned}$$

$$Gini(\text{student}) = 0.745 - 0.32 = \underline{0.425} \text{ Max.}$$

$$Gini(\text{Credit-rating}) = 0.745 - 0.41 = \underline{0.335}$$



age	stuc-R	Income	buys-Comp	age	Income	C-R	buys-Computer
Senior	Fair	Low	Yes	Youth	High	Fair	No
Senior	Excellent	Low	No	Youth	High	Excellent	No
Youth	Fair	Low	Yes	Senior	Medium	Fair	Yes
Senior	Fair	Medium	yes	Youth	Medium	Fair	No
Youth	Excellent	Medium	Yes	Senior	Medium	Excellent	NO

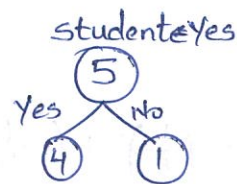
Consider branch, student = yes

15

step 1:

$$Gini(D) = \left[1 - \left(\frac{4}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right]$$

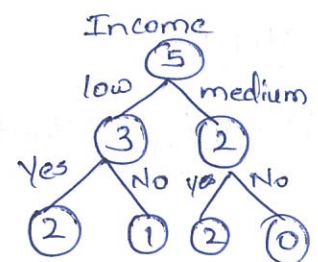
$$= \boxed{0.32}$$



step 2:

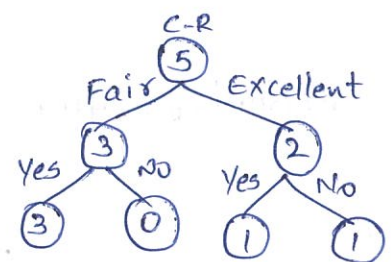
$$i) \underline{Gini_{income \in low}} = \frac{3}{5} \left[1 - \left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] + \frac{2}{5} \left[1 - \left(\frac{2}{2} \right)^2 \right]$$

$$= \boxed{0.2664}$$



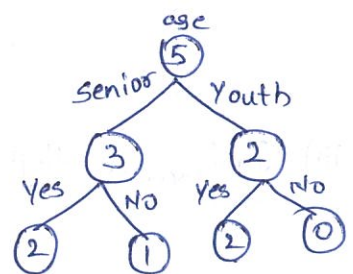
$$ii) \underline{Gini_{C-R \in Excellent}} = \frac{2}{5} \left[1 - \left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] + \frac{3}{5} \left[1 - \left(\frac{3}{3} \right)^2 \right]$$

$$= \boxed{0.2664}$$



$$iii) \underline{Gini_{age \in Senior}} = \frac{3}{5} \left[1 - \left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] + \frac{2}{5} \left[1 - \left(\frac{2}{2} \right)^2 \right]$$

$$= \boxed{0.2664}$$

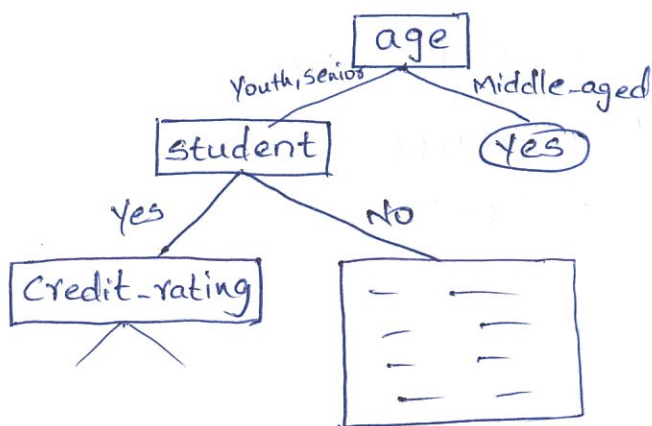


step 3:

$$i) Gini(age) = 0.32 - 0.2664 = \underline{0.0536}$$

$$ii) Gini(Income) = 0.32 - 0.2664 = \underline{0.0536}$$

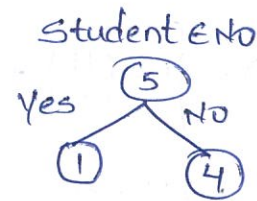
$$iii) Gini(C-R) = 0.32 - 0.2 = \underline{0.1} \text{ Max.}$$



Consider a branch, student = No

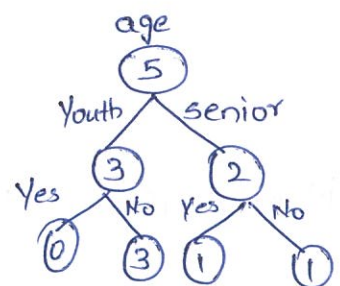
Step 1:

$$\begin{aligned} \text{Gini}(D) &= \left[1 - \left(\frac{1}{5} \right)^2 + \left(\frac{4}{5} \right)^2 \right] \\ &= 1 - 0.68 \\ &= \boxed{0.32} \end{aligned}$$

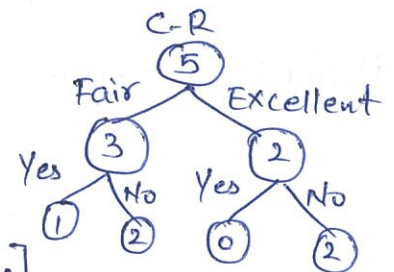


Step 2:

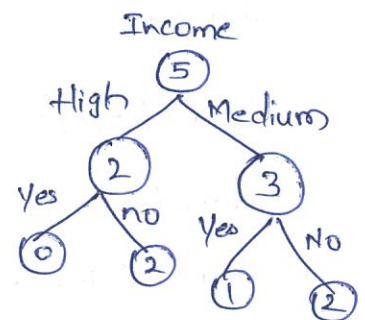
$$\begin{aligned} \text{i) } \text{Gini}_{\text{age} \in \text{Youth}} &= \frac{3}{5} \left[1 - \left(\frac{3}{3} \right)^2 \right] + \frac{2}{5} \left[1 - \left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] \\ &= \boxed{0.2} \end{aligned}$$



$$\begin{aligned} \text{ii) } \text{Gini}_{\text{C-R} \in \text{fair}} &= \frac{3}{5} \left[1 - \left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right] + \frac{2}{5} \left[1 - \left(\frac{2}{2} \right)^2 \right] \\ &= \frac{3}{5} (0.444) \\ &= \boxed{0.2664} \end{aligned}$$



$$\begin{aligned} \text{iii) } \text{Gini}_{\text{Income} \in \text{High}} &= \frac{2}{5} \left[1 - \left(\frac{2}{2} \right)^2 \right] + \frac{3}{5} \left[1 - \left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right] \\ &= \frac{3}{5} (0.444) \\ &= \boxed{0.2664} \end{aligned}$$

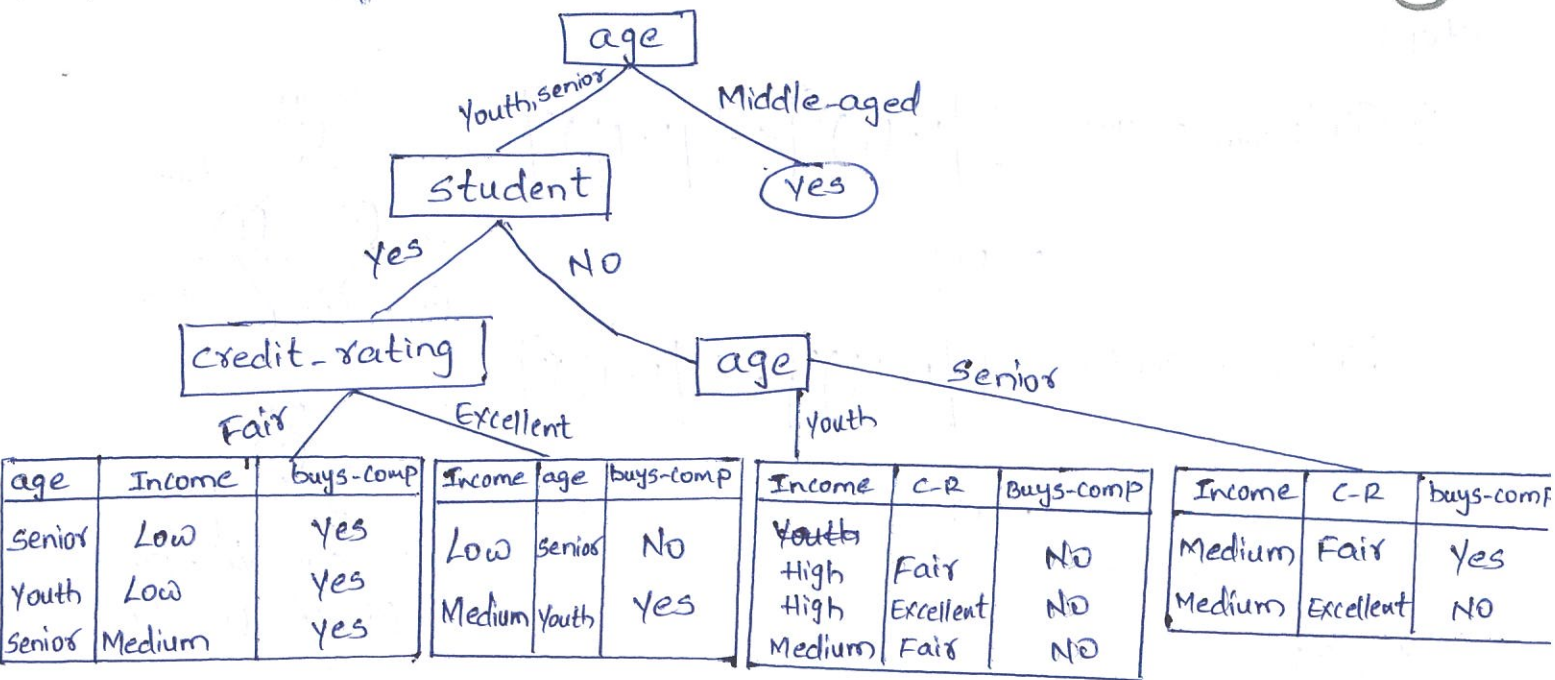


Step 3:

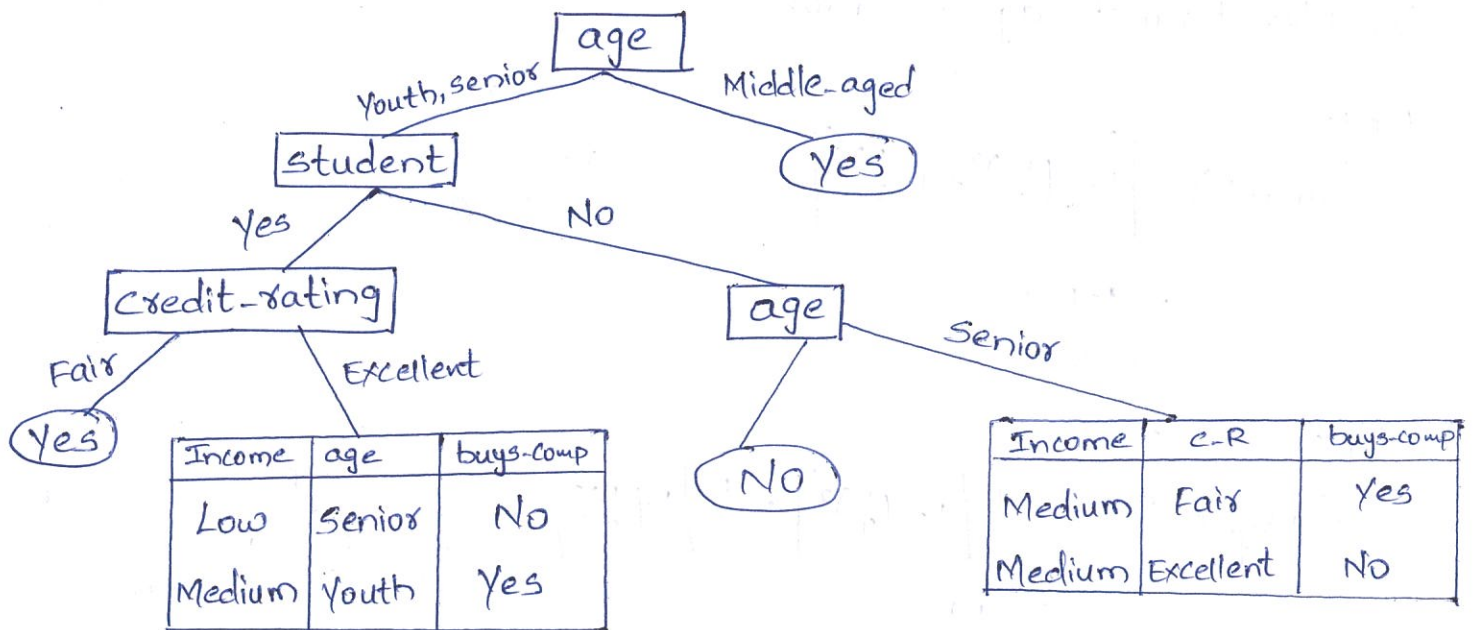
$$\text{i) } \text{Gini}(\text{age}) = 0.32 - 0.2 = \underline{0.1} \text{ — Max}$$

$$\text{ii) } \text{Gini}(\text{C-R}) = 0.32 - 0.2664 = \underline{0.0536}$$

$$\text{iii) } \text{Gini}(\text{Income}) = 0.32 - 0.2664 = \underline{0.0536}$$



* In above tree, C-R ∈ Fair belongs to same class i.e. yes and age ∈ Youth belongs to same class i.e. No.
 ∴ leaf nodes are created for credit-rating and age

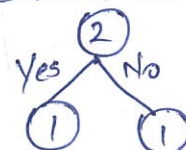


Consider branch, credit-rating = Excellent

step 1:

$$\begin{aligned}
 \text{Gini}(D) &= \left[1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right] \\
 &= 1 - 0.5 \\
 &= \boxed{0.5}
 \end{aligned}$$

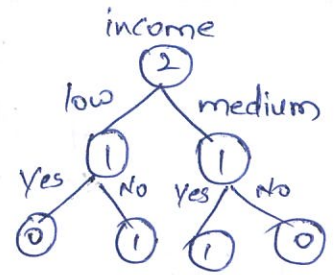
C-R ∈ Excellent



Step 2:

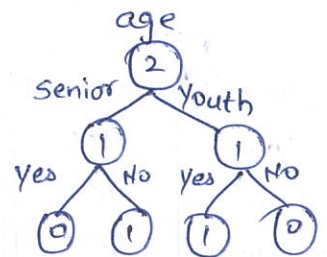
$$i) \underline{Gini_{income \in low}} = \frac{1}{2} \left[1 - \left(\frac{1}{1} \right)^2 \right] + \frac{1}{2} \left[1 - \left(\frac{1}{1} \right)^2 \right]$$

$$= \boxed{0}$$



$$ii) \underline{Gini_{age \in senior}} = \frac{1}{2} \left[1 - \left(\frac{1}{1} \right)^2 \right] + \frac{1}{2} \left[1 - \left(\frac{1}{1} \right)^2 \right]$$

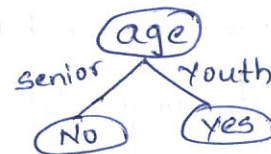
$$= \boxed{0}$$



Step 3:

$$i) Gini(age) = 0.5 - 0 = \underline{0.5}$$

$$ii) Gini(Income) = 0.5 - 0 = \underline{0.5}$$



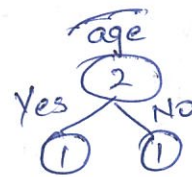
Consider branch, age = senior

step 1:

$$Gini(D) = \left[1 - \left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right]$$

$$= 1 - 0.5$$

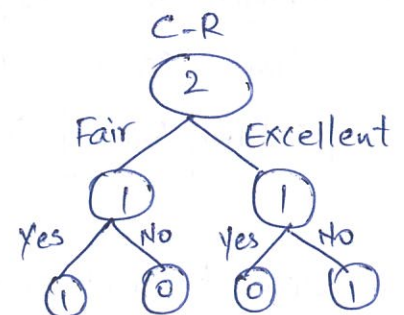
$$= \boxed{0.5}$$



Step 2:

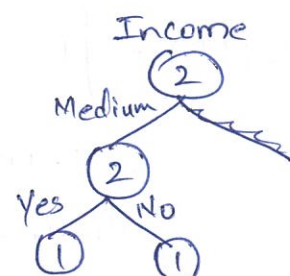
$$i) \underline{Gini_{age \in Fair}^{C-R}} = \frac{1}{2} \left[1 - \left(\frac{1}{2} \right)^2 \right] + \frac{1}{2} \left[1 - \left(\frac{1}{2} \right)^2 \right]$$

$$= \boxed{0}$$



$$ii) \underline{Gini_{income \in Medium}} = \frac{2}{2} \left[1 - \left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right]$$

$$= \boxed{0}$$



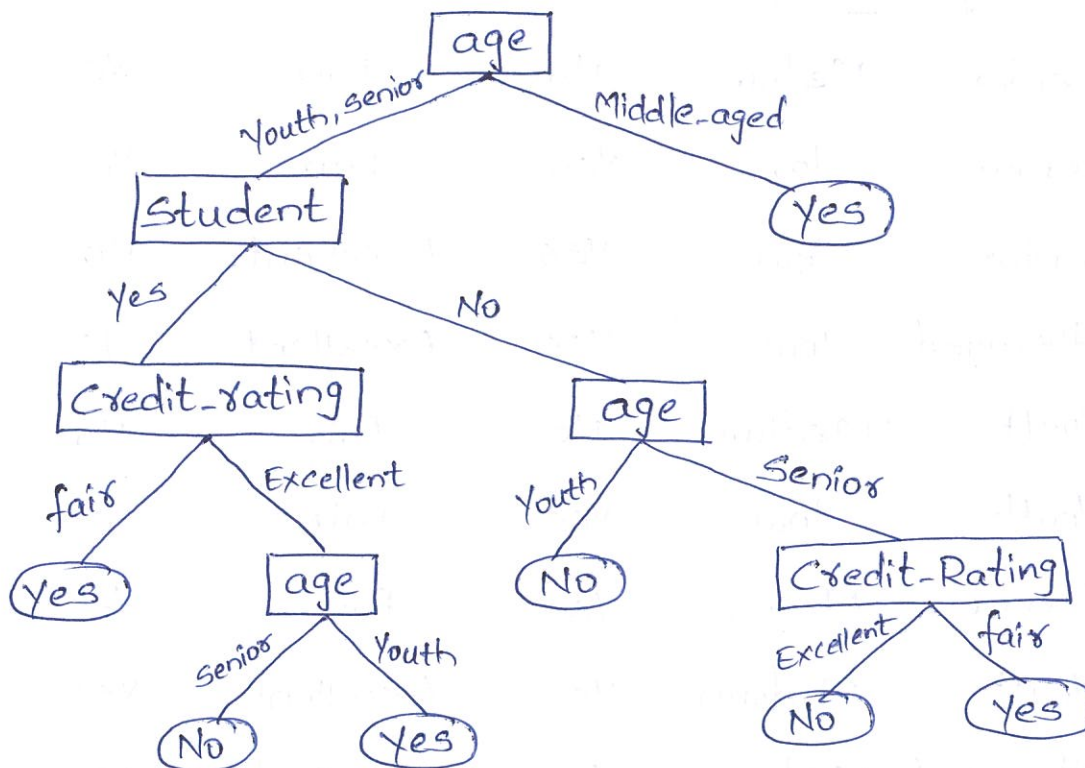
Step 3:

i) $Gini(Income) = 0.5 - 0 = \underline{0.5}$

ii) $Gini(C-R) = 0.5 - 0 = \underline{0.5}$

Credit_Rating	
fair	Excellent
Income	buys-Comp
Medium	yes
Medium	No

* When the value is same for two attributes, any one of the can be considered. The final decision tree is



All Electronics Customer Database with RID.

RID	Age	Income	Student	Credit-rating	class: buys-computer
A	Youth	High	No	Fair	No
B	Youth	High	No	Excellent	No
C	Middle-aged	High	No	Fair	Yes
D	senior	Medium	No	Fair	Yes
E	senior	low	Yes	Fair	Yes
F	senior	low	Yes	Excellent	No
G	Middle-aged	low	Yes	Excellent	Yes
H	Youth	Medium	No	Fair	No
I	Youth	low	Yes	Fair	Yes
J	senior	Medium	Yes	Fair	Yes
K	Youth	Medium	Yes	Excellent	Yes
L	Middle-aged	Medium	No	Excellent	Yes
M	Middle-aged	High	Yes	Fair	Yes
N	senior	Medium	No	Excellent	No

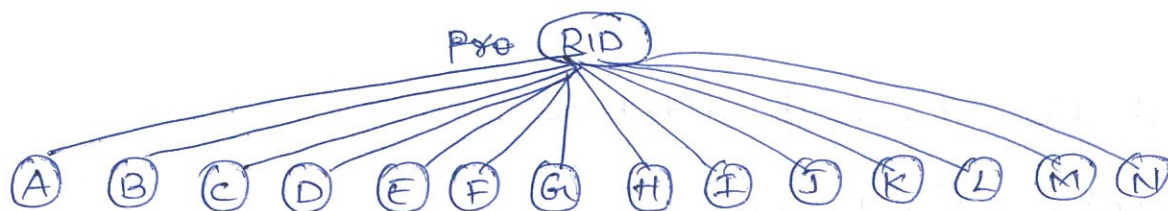
C4.5

* C4.5 is the extension of basic ID₃ algorithm

Drawbacks of ID₃

- 1. In case of unique values in an attribute the tree doesn't grow.

For Ex: Consider ^{RID} product~~ID~~ which has unique values, if we apply a $\text{Info}_A(D)$ in ID₃ we get tree as follows



* The above tree doesn't grow as there are unique values.

* To overcome that drawback C4.5 algorithm is used

C4.5 algorithm procedure.

Step 1: Find $\text{Gain}(D)$ similar procedure done in ID_3

Step 2: Find $\text{SplitInfo}_A(D)$ for each attribute in Dataset D

$$\text{SplitInfo}_A(D) = - \sum_{j=1}^V \frac{|D_j|}{|D|} \log_2 \frac{|D_j|}{|D|}$$

Step 3: Find $\text{GainRatio}_A(D)$ for each attribute in Dataset D .

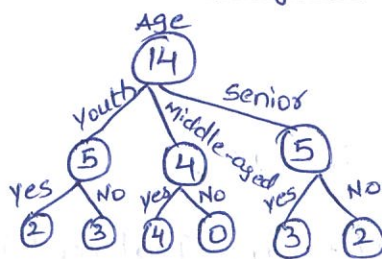
$$\text{GainRatio}(A) = \frac{\text{Gain}(A)}{\text{SplitInfo}(A)}$$

Step 4: After finding above 3 steps pick up the maximum GainRatio and as a splitting attribute

Step 5: Repeat the same procedure until leaf nodes are created.

ProblemStep 1: Find Gain for each attribute

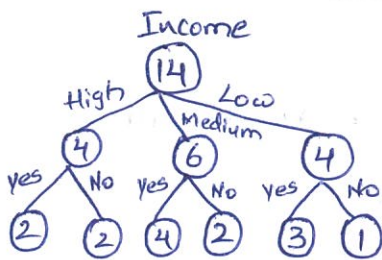
$$\begin{aligned}
 i) \text{ Info}(D) &= - \sum_{i=1}^3 P_i \log_2(P_i) \\
 &= - \frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\
 &= \boxed{0.940}
 \end{aligned}$$



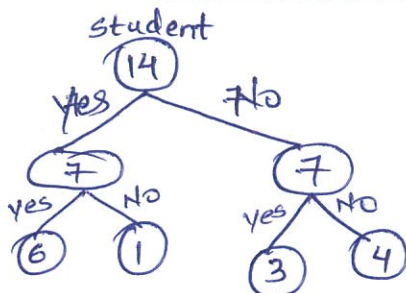
$$\begin{aligned}
 ii) \text{ Info}_{\text{Age}}(D) &= \frac{5}{14} \left(-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right) + \frac{4}{14} \left(-\frac{4}{4} \log_2\left(\frac{4}{4}\right) - 0 \right) + \\
 &\quad \frac{5}{14} \left[-\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right] \\
 &= 0.346 + 0 + 0.346 \\
 &= \boxed{0.692}
 \end{aligned}$$

$$\begin{aligned}
 \cdot \text{ Info}_{\text{RID}}(D) &= \frac{1}{14} \left[-\frac{1}{1} \log_2\left(\frac{1}{1}\right) - 0 \right] + \dots \dots \dots 14 \text{ times} \\
 &= 0 + \dots \dots 14 \text{ times} \\
 &= \boxed{0}
 \end{aligned}$$

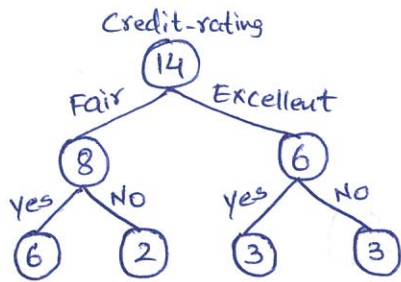
$$\begin{aligned}
 \cdot \text{ Info}_{\text{Income}}(D) &= \frac{4}{14} \left[-\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \right] + \frac{6}{14} \left[-\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \right] \\
 &\quad + \frac{4}{14} \left[-\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \right] \\
 &= 0.285 + 0.393 + 0.231 \\
 &= \boxed{0.909}
 \end{aligned}$$



$$\begin{aligned}
 \cdot \text{ Info}_{\text{student}}(D) &= \frac{7}{14} \left[-\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) \right] + \\
 &\quad \frac{7}{14} \left[-\frac{3}{7} \log_2\left(\frac{3}{7}\right) - \frac{4}{7} \log_2\left(\frac{4}{7}\right) \right] \\
 &= 0.295 + 0.492 \\
 &= \boxed{0.787}
 \end{aligned}$$



$$\text{Info}_{\text{credit-rating}}(D) = \frac{8}{14} \left[-\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) \right] +$$



$$\frac{6}{14} \left[-\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right]$$

$$= 0.463 + 0.428$$

$$= \boxed{0.891}$$

$$\text{iii) Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

$$\text{Gain}(\text{RID}) = 0.940 - 0 = \boxed{0.940}$$

$$\text{Gain}(\text{age}) = 0.940 - 0.692 = \boxed{0.248}$$

$$\text{Gain}(\text{Income}) = 0.940 - 0.909 = \boxed{0.031}$$

$$\text{Gain}(\text{Student}) = 0.940 - 0.787 = \boxed{0.153}$$

$$\text{Gain}(\text{Credit-rating}) = 0.940 - 0.891 = \boxed{0.049}$$

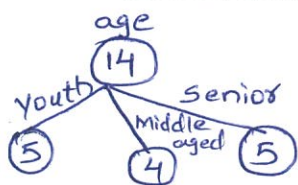
Step 2: Find SplitInfo for each attribute

$$\text{SplitInfo}_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$\text{i) SplitInfo}_{\text{RID}}(D) = - \frac{1}{14} \log_2 \left(\frac{1}{14} \right) - \frac{1}{14} \log_2 \left(\frac{1}{14} \right) \dots \dots 14 \text{ times.}$$

$$= -0.2719 \dots \dots 14 \text{ times} = 3.807$$

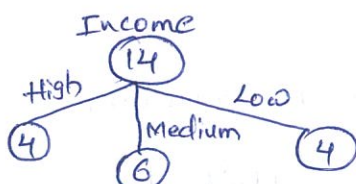
$$\text{ii) SplitInfo}_{\text{age}}(D) = - \frac{5}{14} \log_2 \left(\frac{5}{14} \right) - \frac{4}{14} \log_2 \left(\frac{4}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right)$$



$$= - \left[-0.5305 - 0.5163 - 0.5305 \right]$$

$$= \boxed{1.5773}$$

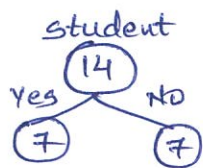
$$\text{iii) SplitInfo}_{\text{Income}}(D) = - \frac{4}{14} \log_2 \left(\frac{4}{14} \right) - \frac{6}{14} \log_2 \left(\frac{6}{14} \right) - \frac{4}{14} \log_2 \left(\frac{4}{14} \right)$$



$$= - \left[-0.5163 - 0.5238 - 0.5163 \right]$$

$$= \boxed{1.5564}$$

$$iv) \text{SplitInfo}_{\text{student}}(D) = -\frac{7}{14} \log_2\left(\frac{7}{14}\right) - \frac{7}{14} \log_2\left(\frac{7}{14}\right)$$



$$= -[-0.5 - 0.5]$$

$$= \boxed{1}$$

$$v) \text{SplitInfo}_{\text{credit-rating}}(D) = -\frac{6}{14} \log_2\left(\frac{6}{14}\right) - \frac{8}{14} \log_2\left(\frac{8}{14}\right)$$

$$= -[-0.5238 - 0.4613]$$

$$= \boxed{0.9851}$$

Step 3: Find GainRatio for each attribute.

$$\text{GainRatio}(A) = \frac{\text{Gain}(A)}{\text{SplitInfo}(A)}$$

$$i) \text{GainRatio}(\text{RID}) = \frac{0.940}{3.807} = \boxed{0.2469}$$

$$ii) \text{GainRatio}(\text{age}) = \frac{0.248}{1.5773} = \boxed{0.1572} \text{ Maximum}$$

$$iii) \text{GainRatio}(\text{Income}) = \frac{0.031}{1.5564} = \boxed{0.0199}$$

$$iv) \text{GainRatio}(\text{student}) = \frac{0.153}{1} = \boxed{0.153}$$

$$v) \text{GainRatio}(\text{credit-rating}) = \frac{0.049}{0.9851} = \boxed{0.0497}$$

* of all the above attributes "age" has a maximum GainRatio so age becomes a splitting attribute

Youth					Senior				
RID	Income	student	C-R	buys-comp	RID	Income	student	C-R	buys-comp
A	High	No	Fair	No	D	Medium	No	Fair	Yes
B	High	No	Excellent	No	E	Low	Yes	Fair	Yes
H	Medium	No	Fair	No	F	Low	Yes	Excellent	No
I	Low	Yes	Fair	Yes	J	Medium	Yes	Fair	Yes
K	Medium	Yes	Excellent	Yes	N	Medium	No	Excellent	No

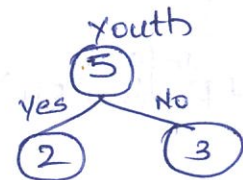
Middle-aged				
RID	Income	student	C-R	buys-computer
C	High	No	Fair	Yes
G	Low	Yes	Excellent	Yes
L	Medium	No	Excellent	Yes
M	High	Yes	Fair	Yes

* A Target variable for middle-aged belong to all "Yes". Hence a leaf node should be created at the end of this branch

Consider branch = Youth

step 1:

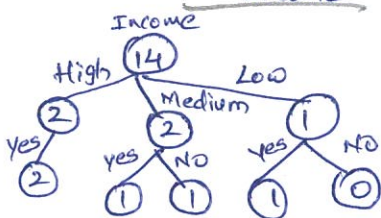
$$i) \text{Info}(D) = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = \boxed{0.970}$$



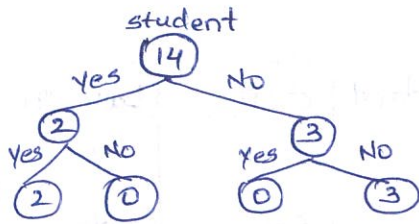
$$ii) \text{Info}_{RID}(D) = \frac{1}{5} \left[-\frac{1}{1} \log_2\left(\frac{1}{1}\right) - 0 \right] + \dots \text{5 times} = \boxed{0}$$

$$\text{Info}_{\text{age}}(D) = \frac{2}{5} \left[-\frac{2}{2} \log_2\left(\frac{2}{2}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right] + \frac{1}{5} \left[-\frac{1}{1} \log_2\left(\frac{1}{1}\right) \right]$$

$$= 0 + 0.4 + 0 = \boxed{0.4}$$

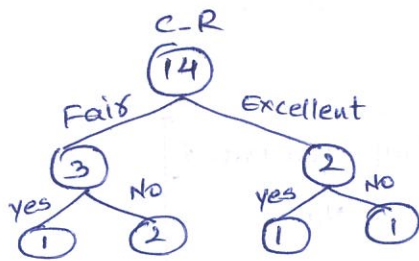


$$\text{Info}_{\text{student}}(D) = \frac{2}{5} \left[-\frac{2}{2} \log_2 \left(\frac{2}{2} \right) \right] + \frac{3}{5} \left[-\frac{3}{3} \log_2 \left(\frac{3}{3} \right) + 0 \right]$$



$$= 0 + 0 = \boxed{0}$$

$$\text{Info}_{\text{credit-rating}}(D) = \frac{3}{5} \left[-\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \right] +$$



$$\frac{2}{5} \left[-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right]$$

$$= 0.550 + 0.4$$

$$= \boxed{0.95}$$

$$\text{iii) Gain(RID)} = 0.970 - 0 = \boxed{0.970}$$

$$\text{Gain(Income)} = 0.970 - 0.4 = \boxed{0.57}$$

$$\text{Gain(student)} = 0.970 - 0 = \boxed{0.970}$$

$$\text{Gain(credit-rating)} = 0.970 - 0.95 = \boxed{0.02}$$

Step 2:

$$\text{i) Split Ratio (RID)} = -\frac{1}{5} \log_2 \left(\frac{1}{5} \right) \dots \dots 5 \text{ times}$$

$$= -0.4643 \dots \dots 5 \text{ times}$$

$$= \boxed{2.3219}$$

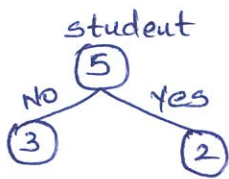
$$\text{ii) Split Ratio (Income)} = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right)$$

$$= 0.528 + 0.528 + 0.46$$

$$= \boxed{1.516}$$



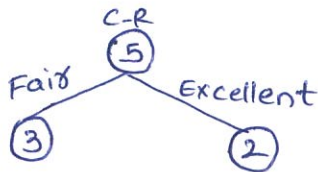
$$\text{iii) SplitInfo(student)} = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$



$$= -[-0.442 - 0.528]$$

$$= \boxed{0.97}$$

$$\text{iv) SplitInfo(Credit-rating)} = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$



$$= -[-0.442 - 0.528]$$

$$= \boxed{0.97}$$

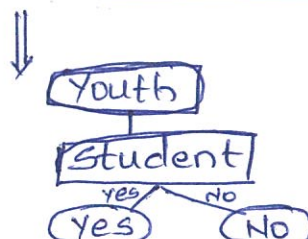
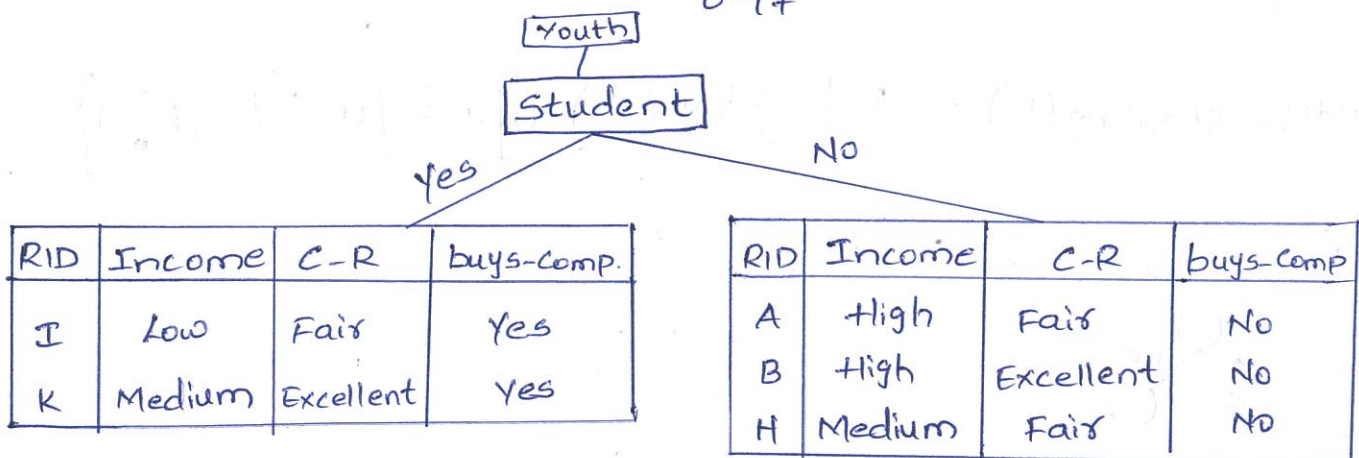
Step: 3

$$\text{i) GainRatio(RID)} = \frac{0.970}{2.3219} = \boxed{0.4177}$$

$$\text{ii) GainRatio(Income)} = \frac{0.57}{1.516} = \boxed{0.3759}$$

$$\text{iii) GainRatio(student)} = \frac{0.970}{0.97} = \boxed{1} \text{ Maximum}$$

$$\text{iv) GainRatio(Credit-rating)} = \frac{0.02}{0.97} = \boxed{0.020}$$

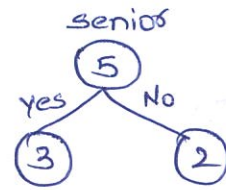


Consider branch = senior

step 1:

$$i) \underline{\text{Info}(D)} = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$

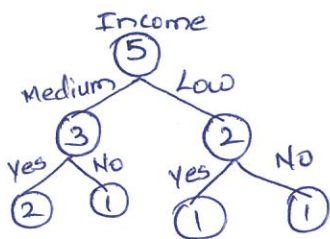
$$= \boxed{0.970}$$



$$ii) \underline{\text{Info}_{RID}(D)} = \frac{1}{5} \left[-\frac{1}{1} \log_2\left(\frac{1}{1}\right) - 0 \right] + \dots \text{ 5 times}$$

$$= \boxed{0}$$

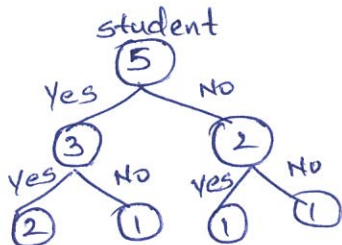
$$\underline{\text{Info}_{\text{Income}}(D)} = \frac{3}{5} \left[-\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$



$$= 0.550 + 0.4$$

$$= \boxed{0.95}$$

$$\underline{\text{Info}_{\text{student}}(D)} = \frac{3}{5} \left[-\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \right] + \frac{2}{5} \left[-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$

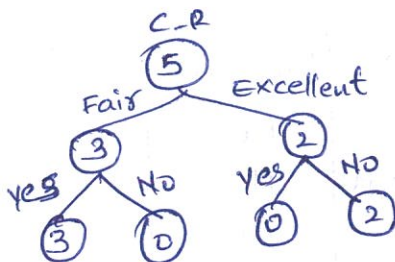


$$= 0.550 + 0.4$$

$$= \boxed{0.95}$$

$$\underline{\text{Info}_{\text{credit-rating}}(D)} = \frac{3}{5} \left[-\frac{3}{3} \log_2\left(\frac{3}{3}\right) - 0 \right] + \frac{2}{5} \left[0 - \frac{2}{2} \log_2\left(\frac{2}{2}\right) \right]$$

$$= \boxed{0}$$



$$\text{iii) Gain (RID)} = 0.970 - 0 = \boxed{0.970}$$

$$\text{Gain (Income)} = 0.970 - 0.95 = \boxed{0.02}$$

$$\text{Gain (Student)} = 0.970 - 0.95 = \boxed{0.02}$$

$$\text{Gain (Credit-Rating)} = 0.970 - 0 = \boxed{0.970}$$

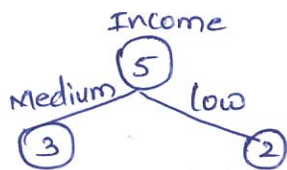
Step 2:

$$\text{i) Split Info (RID)} = -\frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \dots \text{ 5 times}$$

$$= -0.4643 \dots \text{ 5 times}$$

$$= \boxed{2.3219}$$

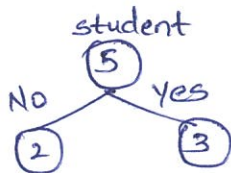
$$\text{ii) Split Info (Income)} = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right)$$



$$= -[-0.442 - 0.528]$$

$$= \boxed{0.97}$$

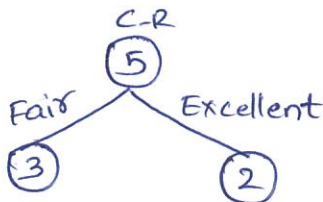
$$\text{iii) Split Info (student)} = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$



$$= -[-0.442 - 0.528]$$

$$= \boxed{0.97}$$

$$\text{iv) Split Info (Credit-Rating)} = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right)$$



$$= -[-0.442 - 0.528]$$

$$= \boxed{0.97}$$

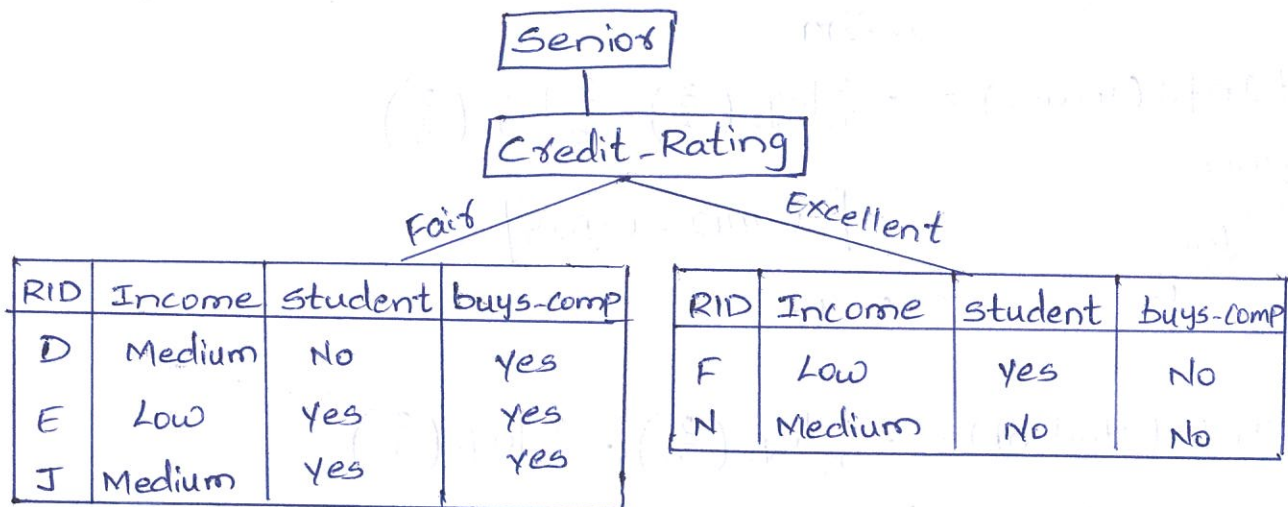
Step 3:

$$i) \text{GainRatio(RID)} = \frac{0.970}{2.3219} = \boxed{0.4177}$$

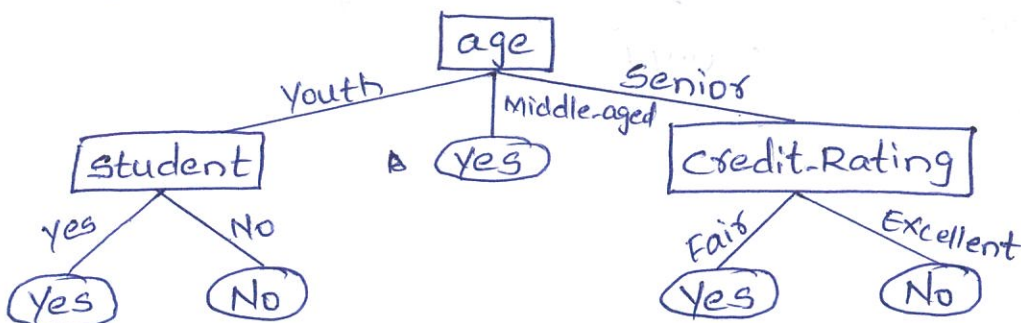
$$ii) \text{GainRatio(Income)} = \frac{0.02}{0.97} = \boxed{0.020}$$

$$iii) \text{GainRatio(student)} = \frac{0.02}{0.97} = \boxed{0.020}$$

$$iv) \text{GainRatio(Credit-rating)} = \frac{0.970}{0.97} = \boxed{1} \text{ Max.}$$



The final decision tree is as follows.



Naive Bayesian classification

- * It is a statistical classifier which performs probabilistic prediction, i.e. predicts class membership probabilities.

Baye's theorem:

$$P(C_i/x) = \frac{P(x/C_i)P(C_i)}{P(x)}$$

Where

$X = (x_1, x_2, \dots, x_n)$ - each tuple

$C = C_1, C_2, \dots, C_m$ (If there are m classes)

$\therefore P(x)$ is Constant for all classes only,

$P(C_i/x) = P(x/C_i)P(x)$ needs to be maximized.

Where, $P(x/C_i) = \prod_{k=1}^n P(x_k/C_i)$

Problem:

Given tuple:

$x: (\overset{x_1}{\text{age}} = \text{youth}, \overset{x_2}{\text{Income}} = \text{medium}, \overset{x_3}{\text{student}} = \text{yes}, \overset{x_4}{\text{Credit-rating}} = \text{fair})$

Now, find probability for each class.

$P(C_1) = \text{buys-Computer} - \text{Yes}$

$$= \frac{9}{14} = \underline{0.6428}$$

$P(C_2) = \text{buys-Computer} - \text{No}$

$$= \frac{5}{14} = \underline{0.357}$$

Now find Probability for each attribute of each class

$$P(x_1/C_1) = \frac{P(x_1 \cap C_1)}{P(C_1)} = \frac{\frac{2}{14}}{\frac{9}{14}} = \underline{0.222}$$

(condition prob.)
= $\frac{\text{No. of possible outcome}}{\text{Total event}}$

$$P(x_1/C_2) = \frac{P(x_1 \cap C_2)}{P(C_2)} = \frac{\frac{3}{14}}{\frac{5}{14}} = \underline{0.6}$$

$$P(x_2/C_1) = \frac{P(x_2 \cap C_1)}{P(C_1)} = \frac{\frac{4}{14}}{\frac{9}{14}} = \underline{0.444}$$

$$P(x_2/C_2) = \frac{P(x_2 \cap C_2)}{P(C_2)} = \frac{\frac{2}{14}}{\frac{5}{14}} = \underline{0.4}$$

$$P(x_3/c_1) = \frac{P(x_3 \cap c_1)}{P(c_1)} = \frac{\frac{6}{14}}{\frac{9}{14}} = 0.666$$

$$P(x_4/c_1) = \frac{P(x_4 \cap c_1)}{P(c_1)} = \frac{\frac{6}{14}}{\frac{9}{14}} = 0.666$$

Now multiply all probabilities of attribute of class-yes

$$P(x/c_1) = 0.222 \times 0.444 \times 0.666 \times 0.666 = 0.0437$$

$$P(x/c_1) = 0.0437$$

$$\begin{aligned} P(c_1/x) &= P(x/c_1) \cdot P(c_1) \\ &= 0.0437 \times 0.6428 \\ &= 0.028 \text{ maximum.} \end{aligned}$$

$$P(x_3/c_2) = \frac{P(x_3 \cap c_2)}{P(c_2)} = \frac{\frac{1}{14}}{\frac{5}{14}} = 0.2$$

$$P(x_4/c_2) = \frac{P(x_4 \cap c_2)}{P(c_2)} = \frac{\frac{2}{14}}{\frac{5}{14}} = 0.4$$

Multiply all probabilities of attribute of class-No

$$P(x/c_2) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.0192$$

$$P(x/c_2) = 0.0192$$

$$\begin{aligned} P(c_2/x) &= P(x/c_2) \cdot P(c_2) \\ &= 0.0192 \times 0.357 \\ &= 0.00685 \end{aligned}$$

$\therefore$ The maximum probability is for buys-Computer = yes

Hence,

The Bayesian classifier predicts yes for given tuple x .

K-Nearest Neighbours

- * It is a classification strategy that is an example of a "lazy learner"
- * A lazy learner do not require building a model with a training set rather it uses training set directly to classify an input when an input is given.

• Now Consider a AllElectronics Customer Database set

Problem:

Given tuple

x : (age = Youth, Income = Medium, Student = Yes, Credit-rating = fair)

- In case of Categorical Values, simple matching measure is used
- We compare given tuple and each individual tuple in dataset
- If attribute Matches, it is 1 otherwise 0.

Tuples		Weights
1	$1 + 0 + 0 + 1$	$= 2$
2	$1 + 0 + 0 + 0$	$= 1$
3	$0 + 0 + 0 + 1$	$= 1$
4	$0 + 1 + 0 + 1$	$= 2$
5	$0 + 0 + 1 + 1$	$= 2$
6	$0 + 0 + 1 + 0$	$= 1$
7	$0 + 0 + 1 + 0$	$= 1$
8	$1 + 1 + 0 + 1$	$= 3$
9	$1 + 0 + 1 + 1$	$= 3$
10	$0 + 1 + 1 + 1$	$= 3$
11	$1 + 0 + 1 + 0$	$= 2$
12	$0 + 1 + 0 + 0$	$= 1$
13	$0 + 0 + 1 + 1$	$= 2$
14	$0 + 1 + 0 + 0$	$= 1$

Arrange the weights in Ascending order

weights : 1 1 1 1 1 1 2 2 2 2 2 3 3 3
 Tuples : 2 3 6 7 12 14 1 4 5 11 13 8 9 10
 w.r.t weights

Now Assume K value. Before assuming K value following Conditions should satisfy

* $K \approx \sqrt{n}$ where n = no. of tuples

* The n values should not be the multiples of K .

For Ex: If $K=7$ where $n=14$

Here 14 is multiple of 7 so $K=7$ should not be considered.

* The K value should be an odd number so that there will no tie between classes.

In our dataset $n=14$

$$\boxed{K \approx \sqrt{n}} \Rightarrow K \approx \sqrt{14}$$

≈ 3.74 (consider next closest odd no.)

$$\therefore K=5$$

Now take first five tuples which are in Ascending order

Tuples : 2 3 6 7 12

Corresponding : No Yes No Yes Yes
 classes

The maximum voting of a class from a sample is the class for a given tuple.

Here

$$\text{class_No} = 2$$

$$\text{class_Yes} = 3 \text{ (Max.)}$$

$\therefore$ The KNN predicts "yes" for given tuple x .